

## 12.7 Products Entry by Entry

Matrix multiplication is not the only way to multiply two matrices of the same size. There is a much simpler product — multiply the entries in matching positions — which is useless for composing linear maps and extremely useful everywhere else. This section introduces it and proves the one theorem that makes it interesting: it preserves positivity. A determinant inequality follows, and it contains Hadamard’s inequality of Section 6 as a special case.

Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$ .

### 12.7.1 The entrywise product

Two matrices of the same shape can be multiplied the way two lists of numbers are: position by position. The operation has no interpretation in terms of linear maps, which is one reason it looks like a curiosity at first sight, and it is written with a small circle to keep it away from ordinary multiplication.

*The Hadamard product multiplies matrices the way a spreadsheet would.*

#### Definition 12.7.1 — Hadamard Product

Let  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$ . Their **Hadamard product**, or **entrywise product**, is the matrix  $\mathbf{A} \circ \mathbf{B} \in M_{m \times n}(F)$  with

$$(\mathbf{A} \circ \mathbf{B})_{ij} = a_{ij}b_{ij} \quad \text{for all } i, j.$$

Both factors must have **the same shape**, and the result has that shape too; there is no rule like “columns of the first match rows of the second”. The operation is commutative and associative, and it is bilinear —  $(\mathbf{A} + \mathbf{A}') \circ \mathbf{B} = \mathbf{A} \circ \mathbf{B} + \mathbf{A}' \circ \mathbf{B}$  and  $(c\mathbf{A}) \circ \mathbf{B} = c(\mathbf{A} \circ \mathbf{B})$  — because each of these is a statement about numbers, checked one entry at a time. Its identity element is the all-ones matrix  $\mathbf{J}$ , not  $\mathbf{I}$ .

**Examples.**

- $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \circ \begin{pmatrix} 5 & 0 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ -3 & 8 \end{pmatrix}$ .
- $\mathbf{A} \circ \mathbf{I} = \text{diag}(a_{11}, \dots, a_{nn})$ : multiplying entrywise by the identity **erases** everything off the diagonal, where ordinary multiplication by  $\mathbf{I}$  changes nothing.
- $\mathbf{A} \circ \mathbf{J} = \mathbf{A}$ , so  $\mathbf{J}$  is the identity for  $\circ$ .
- For two diagonal matrices,  $\mathbf{D} \circ \mathbf{D}' = \mathbf{DD}'$ : both products are  $\text{diag}(d_1d'_1, \dots, d_nd'_n)$ . For matrices in general the two have nothing to do with each other, though they can coincide by accident:  $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  have equal Hadamard and matrix products without either being diagonal.
- For column vectors  $\mathbf{u}, \mathbf{w} \in F^n$ , regarded as  $n \times 1$  matrices,  $\mathbf{u} \circ \mathbf{w} = (u_1w_1, \dots, u_nw_n)$ .

**Non-example by minimal change.** Take  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{B} = \mathbf{A}$ . Then  $\mathbf{AB} = \mathbf{I}$ , but  $\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \circ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . The two products differ, and they differ in the most basic way:  $\mathbf{A}$  is its own inverse and its own Hadamard square.

*Warning.*  $\mathbf{A} \circ \mathbf{B}$  is not  $\mathbf{AB}$ , and almost nothing carries over. Two invertible matrices can have a singular entrywise product:

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \circ \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix},$$

where the two factors have determinants  $-3$  and  $3$  and the product has determinant  $0$ . There is no formula for  $\det(\mathbf{A} \circ \mathbf{B})$  in terms of  $\det \mathbf{A}$  and  $\det \mathbf{B}$ , no relation between the eigenvalues, and no rule  $(\mathbf{A} \circ \mathbf{B})^{-1} = \mathbf{A}^{-1} \circ \mathbf{B}^{-1}$ . The one structural property that does survive is the subject of this section, and it is a genuine surprise.

The name is the same one that Section 6 attached to a determinant inequality. That is a coincidence of a busy mathematician, not of the mathematics: Hadamard's inequality and the Hadamard product are unrelated statements, and this section will nonetheless prove one from the other.

## 12.7.2 The Schur product theorem

Positivity is a statement about the quadratic form  $\mathbf{x}^* \mathbf{A} \mathbf{x}$ , and the quadratic form of  $\mathbf{A} \circ \mathbf{B}$  has no visible relation to those of  $\mathbf{A}$  and  $\mathbf{B}$ . So the following theorem cannot be proved by staring at the form. What works is to break both matrices into the simplest positive pieces there are — rank-one matrices  $\mathbf{u}\mathbf{u}^*$  — and to notice that the entrywise product of two such pieces is again one.

### Lemma 12.7.1 — Entrywise product of two rank-one positive matrices

Let  $\mathbf{u}, \mathbf{w} \in F^n$ . Then

$$(\mathbf{u}\mathbf{u}^*) \circ (\mathbf{w}\mathbf{w}^*) = (\mathbf{u} \circ \mathbf{w})(\mathbf{u} \circ \mathbf{w})^*.$$

*Proof.* Compare the  $(i, j)$  entries. The  $(i, j)$  entry of  $\mathbf{u}\mathbf{u}^*$  is  $u_i \overline{u_j}$  and that of  $\mathbf{w}\mathbf{w}^*$  is  $w_i \overline{w_j}$ , so the left-hand side has  $(i, j)$  entry

$$u_i \overline{u_j} w_i \overline{w_j} = (u_i w_i) \overline{(u_j w_j)},$$

the rearrangement using commutativity of  $F$  and  $\overline{u_j} \overline{w_j} = \overline{u_j w_j}$ . The vector  $\mathbf{u} \circ \mathbf{w}$  has  $i$ -th entry  $u_i w_i$ , so the right-hand side has  $(i, j)$  entry  $(u_i w_i) \overline{(u_j w_j)}$  as well. The two matrices agree entry by entry. ■

That one line is the whole theorem; everything else is bookkeeping over the spectral decompositions.

**Theorem 12.7.2 — Schur Product Theorem**

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ .

- (a) If  $\mathbf{A} \succeq 0$  and  $\mathbf{B} \succeq 0$ , then  $\mathbf{A} \circ \mathbf{B} \succeq 0$ .
- (b) If  $\mathbf{A} \succ 0$  and  $\mathbf{B} \succ 0$ , then  $\mathbf{A} \circ \mathbf{B} \succ 0$ .

*Idea.* For (a): the spectral theorem writes  $\mathbf{A} = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^*$  and  $\mathbf{B} = \sum_j \mu_j \mathbf{w}_j \mathbf{w}_j^*$  with all  $\lambda_i, \mu_j \geq 0$ . Expand  $\mathbf{A} \circ \mathbf{B}$  by bilinearity into  $n^2$  terms; by Lemma 12.7.1 each term is a non-negative multiple of a matrix  $\mathbf{z}\mathbf{z}^*$ , and those are positive semidefinite. A sum of positive semidefinite matrices is positive semidefinite, so we are done.

For (b), do not redo the argument with strict inequalities: the vectors  $\mathbf{u}_i \circ \mathbf{w}_j$  need not span, and the bookkeeping becomes unpleasant. Instead peel a multiple of the identity off  $\mathbf{B}$ . If  $\varepsilon > 0$  is the smallest eigenvalue of  $\mathbf{B}$ , then  $\mathbf{B} = (\mathbf{B} - \varepsilon \mathbf{I}) + \varepsilon \mathbf{I}$  with  $\mathbf{B} - \varepsilon \mathbf{I} \succeq 0$ , so

$$\mathbf{A} \circ \mathbf{B} = \mathbf{A} \circ (\mathbf{B} - \varepsilon \mathbf{I}) + \varepsilon (\mathbf{A} \circ \mathbf{I}),$$

and  $\mathbf{A} \circ \mathbf{I}$  is the diagonal matrix of the  $a_{ii}$ , which are strictly positive. Part (a) handles the first term, and a positive definite matrix plus a positive semidefinite one is positive definite.

*Proof.*

- (a) Both  $\mathbf{A}$  and  $\mathbf{B}$  are Hermitian, so each has an orthonormal basis of eigenvectors. Let  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  be one for  $\mathbf{A}$ , with  $\mathbf{A}\mathbf{u}_i = \lambda_i \mathbf{u}_i$ , and let  $\mathbf{U}$  be the unitary matrix with these columns, so that  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$  by Corollary 11.4.3 over  $\mathbb{C}$  and Corollary 11.5.2 over  $\mathbb{R}$ . The columns of  $\mathbf{U}\mathbf{D}$  are  $\lambda_i \mathbf{u}_i$  and the rows of  $\mathbf{U}^*$  are  $\mathbf{u}_i^*$ , so Corollary 7.1.3 gives

$$\mathbf{A} = \sum_{i=1}^n \lambda_i \mathbf{u}_i \mathbf{u}_i^*,$$

which over  $\mathbb{R}$  is Corollary 11.5.4. In the same way  $\mathbf{B} = \sum_j \mu_j \mathbf{w}_j \mathbf{w}_j^*$ . Every  $\lambda_i \geq 0$  and every  $\mu_j \geq 0$ , by Theorem 12.1.4 (b).

The Hadamard product is bilinear, so expanding both sums,

$$\mathbf{A} \circ \mathbf{B} = \sum_{i=1}^n \sum_{j=1}^n \lambda_i \mu_j [(\mathbf{u}_i \mathbf{u}_i^*) \circ (\mathbf{w}_j \mathbf{w}_j^*)] = \sum_{i,j} \lambda_i \mu_j \mathbf{z}_{ij} \mathbf{z}_{ij}^*,$$

where  $\mathbf{z}_{ij} = \mathbf{u}_i \circ \mathbf{w}_j$  and the second equality is Lemma 12.7.1. Each  $\mathbf{z}_{ij} \mathbf{z}_{ij}^*$  is positive semidefinite by Example (Five Positive Matrices) (b), and each coefficient  $\lambda_i \mu_j$  is a non-negative real number. A non-negative combination of positive semidefinite matrices is positive semidefinite (Exercise (C2: The cone of positive matrices) (a)), so  $\mathbf{A} \circ \mathbf{B} \succeq 0$ .

- (b) Let  $\varepsilon > 0$  be the smallest eigenvalue of  $\mathbf{B}$ , which is positive by Theorem 12.1.5 (b). The matrix  $\mathbf{B} - \varepsilon \mathbf{I}$  is Hermitian with eigenvalues  $\mu_j - \varepsilon \geq 0$ , so  $\mathbf{B} - \varepsilon \mathbf{I} \succeq 0$  by Theorem 12.1.4 (b). Since  $\circ$  is bilinear and  $\mathbf{A} \circ \mathbf{I} = \text{diag}(a_{11}, \dots, a_{nn})$ ,

$$\mathbf{A} \circ \mathbf{B} = \mathbf{A} \circ (\mathbf{B} - \varepsilon \mathbf{I}) + \varepsilon \text{diag}(a_{11}, \dots, a_{nn}).$$

The first term is positive semidefinite by (a). In the second,  $a_{ii} = \langle \mathbf{A}\mathbf{e}_i, \mathbf{e}_i \rangle > 0$  because  $\mathbf{A} \succ 0$  and  $\mathbf{e}_i \neq \mathbf{0}$ , so  $\varepsilon \operatorname{diag}(a_{11}, \dots, a_{nn}) \succ 0$  by Example (Five Positive Matrices) (a). For  $\mathbf{x} \neq \mathbf{0}$ , the quadratic form of the sum is  $\geq 0$  plus  $> 0$ , hence  $> 0$ . So  $\mathbf{A} \circ \mathbf{B} \succ 0$ , and the theorem is proved. ■

The theorem is a genuine addition to the list of operations that preserve positivity. Sums and non-negative multiples were easy (Exercise (C2: The cone of positive matrices)), congruence was easy (Proposition 12.1.2), and the ordinary product is not on the list at all, since  $\mathbf{AB}$  need not even be Hermitian. The entrywise product is on the list, and nothing in the definition of positivity suggests it should be.

**Example. ( $\mathbf{A} \ 3 \times 3$  Entrywise Product)**

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 3 \end{pmatrix}.$$

Check that  $\mathbf{A} \succ 0$  and  $\mathbf{B} \succ 0$ , compute  $\mathbf{A} \circ \mathbf{B}$ , and verify directly that it is positive definite.

*Solution.* The leading principal minors of  $\mathbf{A}$  are 2,  $4 - 1 = 3$  and, expanding along the first row,  $2(4 - 1) - 1(2 - 0) = 4$ . All are positive, so  $\mathbf{A} \succ 0$  by Sylvester's criterion (Theorem 12.1.5 (d)).

Those of  $\mathbf{B}$  are 1,  $2 - 1 = 1$  and  $1(6 - 1) + 1(-3 + 1) + 1(1 - 2) = 5 - 2 - 1 = 2$ . All positive, so  $\mathbf{B} \succ 0$ .

Entry by entry,

$$\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 6 \end{pmatrix},$$

whose leading principal minors are 2,  $8 - 1 = 7$  and  $2(24 - 1) + 1(-6 - 0) = 46 - 6 = 40$ . All positive, so  $\mathbf{A} \circ \mathbf{B} \succ 0$ , as Theorem 12.7.2 (b) promised. Note that the eigenvalues of  $\mathbf{A} \circ \mathbf{B}$  have nothing to do with those of  $\mathbf{A}$  and  $\mathbf{B}$ , and that  $\det(\mathbf{A} \circ \mathbf{B}) = 40$  is much larger than  $\det \mathbf{A} \det \mathbf{B} = 4 \cdot 2 = 8$ . The next theorem explains part of that: it forces  $\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{A} \prod_i b_{ii} = 24$ . Getting down to the product  $\det \mathbf{A} \det \mathbf{B}$  needs Hadamard's inequality as well, which is Exercise (C2: The product of the determinants).

*Quick check.* Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ . Compute  $\mathbf{A} \circ \mathbf{B}$  and decide whether it is positive definite. Does this contradict Theorem 12.7.2 (b)?

*Solution.*  $\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ , with leading principal minors 2 and 3, so it is positive definite. There is no contradiction: Theorem 12.7.2 (b) is an implication, not an equivalence. Here  $\mathbf{A}$  is only positive semidefinite ( $\mathbf{A} = \mathbf{1}\mathbf{1}^\top$ , so  $\det \mathbf{A} = 0$ ), and the theorem's hypothesis

fails — but the conclusion happens to hold anyway. Part (a) is what applies, and it correctly predicts  $\mathbf{A} \circ \mathbf{B} \succeq 0$ .

### 12.7.3 Oppenheim's inequality

Hadamard's inequality of Section 6 compared  $\det \mathbf{A}$  with the product of the diagonal entries of  $\mathbf{A}$ . Oppenheim's inequality does the same for a Hadamard product, and it contains Hadamard's as the special case  $\mathbf{B} = \mathbf{I}$ .

#### Theorem 12.7.3 — Oppenheim's Inequality

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{A} \succeq 0$  and  $\mathbf{B} \succeq 0$ . Then

$$\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{A} \cdot b_{11} b_{22} \cdots b_{nn}.$$

*Idea.* Induct on  $n$ , taking both matrices positive definite first. Split off the  $(1, 1)$  entry of each: the Schur complements  $\mathbf{A}/\alpha$  and  $\mathbf{B}/\beta$  are positive definite of size  $n - 1$ , and a short computation, whose only real step is Lemma 12.7.1 again, shows that the Schur complement of the corner in  $\mathbf{A} \circ \mathbf{B}$  is

$$(\mathbf{A} \circ \mathbf{B})/(\alpha\beta) = (\mathbf{A}/\alpha) \circ \mathbf{B}' + \frac{1}{\alpha} (\mathbf{a}\mathbf{a}^*) \circ (\mathbf{B}/\beta),$$

a positive definite matrix plus a positive semidefinite one. So its determinant is at least that of the first term, by Lemma 12.6.2, and the inductive hypothesis handles that. The semidefinite case follows by replacing  $\mathbf{A}$  and  $\mathbf{B}$  with  $\mathbf{A} + t\mathbf{I}$  and  $\mathbf{B} + t\mathbf{I}$  and letting  $t$  shrink to 0.

*Proof. Case 1:  $\mathbf{A} \succ 0$  and  $\mathbf{B} \succ 0$ .* Induct on  $n$ . For  $n = 1$  both sides are  $a_{11}b_{11}$ .

Let  $n \geq 2$ , and split both matrices with a  $1 \times (n - 1)$  partition:

$$\mathbf{A} = \begin{pmatrix} \alpha & \mathbf{a}^* \\ \mathbf{a} & \mathbf{A}' \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} \beta & \mathbf{b}^* \\ \mathbf{b} & \mathbf{B}' \end{pmatrix},$$

with  $\alpha = a_{11} > 0$ ,  $\beta = b_{11} > 0$  and  $\mathbf{a}, \mathbf{b} \in F^{n-1}$ . The Schur complements are

$$\mathbf{A}/\alpha = \mathbf{A}' - \frac{1}{\alpha} \mathbf{a}\mathbf{a}^*, \quad \mathbf{B}/\beta = \mathbf{B}' - \frac{1}{\beta} \mathbf{b}\mathbf{b}^*,$$

and both are positive definite by Theorem 12.6.1 (b). Since conjugation acts entrywise,  $\mathbf{a}^* \circ \mathbf{b}^* = (\mathbf{a} \circ \mathbf{b})^*$ , so

$$\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} \alpha\beta & (\mathbf{a} \circ \mathbf{b})^* \\ \mathbf{a} \circ \mathbf{b} & \mathbf{A}' \circ \mathbf{B}' \end{pmatrix},$$

and  $\alpha\beta > 0$ , so this matrix too has a Schur complement at its corner:

$$(\mathbf{A} \circ \mathbf{B})/(\alpha\beta) = \mathbf{A}' \circ \mathbf{B}' - \frac{1}{\alpha\beta} (\mathbf{a} \circ \mathbf{b})(\mathbf{a} \circ \mathbf{b})^*.$$

By [Lemma 12.7.1](#) the subtracted term is  $\frac{1}{\alpha\beta}(\mathbf{a}\mathbf{a}^*) \circ (\mathbf{b}\mathbf{b}^*)$ . Substituting  $\mathbf{A}' = \mathbf{A}/\alpha + \frac{1}{\alpha}\mathbf{a}\mathbf{a}^*$  and expanding by bilinearity,

$$\begin{aligned} \mathbf{A}' \circ \mathbf{B}' &= (\mathbf{A}/\alpha) \circ \mathbf{B}' + \frac{1}{\alpha} (\mathbf{a}\mathbf{a}^*) \circ \mathbf{B}' \\ &= (\mathbf{A}/\alpha) \circ \mathbf{B}' + \frac{1}{\alpha} (\mathbf{a}\mathbf{a}^*) \circ \left( \mathbf{B}/\beta + \frac{1}{\beta} \mathbf{b}\mathbf{b}^* \right), \end{aligned}$$

and the last term of this is exactly  $\frac{1}{\alpha\beta}(\mathbf{a}\mathbf{a}^*) \circ (\mathbf{b}\mathbf{b}^*)$ . It therefore cancels, leaving

$$(\mathbf{A} \circ \mathbf{B})/(\alpha\beta) = \underbrace{(\mathbf{A}/\alpha) \circ \mathbf{B}'}_{\succ 0} + \underbrace{\frac{1}{\alpha} (\mathbf{a}\mathbf{a}^*) \circ (\mathbf{B}/\beta)}_{\succeq 0}. \quad (*)$$

The first term is positive definite by [Theorem 12.7.2](#) (b), since  $\mathbf{A}/\alpha \succ 0$  and  $\mathbf{B}' \succ 0$  (the latter by Theorem 12.6.1 (a) applied to  $\mathbf{B}$ ). The second is positive semidefinite by [Theorem 12.7.2](#) (a), since  $\mathbf{a}\mathbf{a}^* \succeq 0$  by Example (Five Positive Matrices) (b) and  $\mathbf{B}/\beta \succ 0$ , and  $1/\alpha > 0$ .

Now compute. By Theorem 7.3.2 applied to  $\mathbf{A} \circ \mathbf{B}$ , then (\*) with Lemma 12.6.2, then the inductive hypothesis applied to the pair  $\mathbf{A}/\alpha$  and  $\mathbf{B}'$ :

$$\begin{aligned} \det(\mathbf{A} \circ \mathbf{B}) &= \alpha\beta \cdot \det[(\mathbf{A} \circ \mathbf{B})/(\alpha\beta)] \\ &\geq \alpha\beta \cdot \det[(\mathbf{A}/\alpha) \circ \mathbf{B}'] \\ &\geq \alpha\beta \cdot \det(\mathbf{A}/\alpha) \cdot b_{22} \cdots b_{nn}. \end{aligned}$$

By Theorem 7.3.2 again, now for  $\mathbf{A}$ , we have  $\det(\mathbf{A}/\alpha) = \det \mathbf{A}/\alpha$ . Substituting and canceling  $\alpha$ ,

$$\det(\mathbf{A} \circ \mathbf{B}) \geq \beta \det \mathbf{A} \cdot b_{22} \cdots b_{nn} = \det \mathbf{A} \cdot b_{11} b_{22} \cdots b_{nn},$$

which completes the induction.

*Case 2:  $\mathbf{A} \succeq 0$  and  $\mathbf{B} \succeq 0$ .* For real  $t > 0$ , the matrices  $\mathbf{A} + t\mathbf{I}$  and  $\mathbf{B} + t\mathbf{I}$  are positive definite, as in the proof of Corollary 12.6.3. Case 1 gives

$$\det[(\mathbf{A} + t\mathbf{I}) \circ (\mathbf{B} + t\mathbf{I})] \geq \det(\mathbf{A} + t\mathbf{I}) \cdot \prod_{i=1}^n (b_{ii} + t).$$

Every entry of  $(\mathbf{A} + t\mathbf{I}) \circ (\mathbf{B} + t\mathbf{I})$  is a polynomial in  $t$ , so both sides are polynomial, hence continuous, functions of  $t$  (Theorem 6.4.3). Letting  $t$  decrease to 0 preserves the inequality and gives the statement. This proves the theorem. ■

*Remark.* Taking  $\mathbf{B} = \mathbf{I}$  recovers Hadamard's inequality. Indeed  $\mathbf{A} \circ \mathbf{I} = \text{diag}(a_{11}, \dots, a_{nn})$  and  $\prod_i b_{ii} = 1$ , so the conclusion reads  $a_{11} \cdots a_{nn} \geq \det \mathbf{A}$ , which is Theorem 12.6.4 (a). Swapping the roles of  $\mathbf{A}$  and  $\mathbf{B}$  gives the companion bound  $\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{B} \cdot \prod_i a_{ii}$ , and combining either one with Hadamard's inequality gives  $\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{A} \det \mathbf{B}$ ; that last consequence is [Exercise \(C2: The product of the determinants\)](#).

## 12.7.4 Where this gets used

The natural home of the Hadamard product is data. A covariance matrix is a Gram matrix (Theorem 10.2.5), hence positive semidefinite, and rescaling each variable to unit variance turns it into a **correlation matrix**: positive semidefinite with every diagonal entry equal to 1. The Schur product theorem says that the entrywise product of two correlation matrices is another one — the diagonal entries stay  $1 \cdot 1 = 1$  — so one can damp the correlations of one model by those of another and still have a legitimate covariance structure. Nothing in the definition of “positive semidefinite” makes that obvious, and it is the reason the theorem is quoted far outside linear algebra.

*Warning.* **Entrywise products preserve positivity; entrywise inverses do not.** Let  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ , which is positive definite, with leading principal minors 2 and 3. Replacing each entry by its reciprocal gives  $\begin{pmatrix} 1/2 & 1 \\ 1 & 1/2 \end{pmatrix}$ , whose determinant is  $\frac{1}{4} - 1 = -\frac{3}{4} < 0$ : not positive semidefinite. The entrywise reciprocal is not the Hadamard inverse of anything useful, and the matrix inverse  $\mathbf{A}^{-1}$ , which **is** positive definite (Exercise (B3: The inverse of a positive definite matrix)), has nothing to do with it.

## 12.7.5 Exercises

### 12.7.5.1 A. Check your understanding

#### Exercise A1.

- Define  $\mathbf{A} \circ \mathbf{B}$ , and say what shapes the two factors must have.
- What is  $\mathbf{A} \circ \mathbf{I}$ ? What is  $\mathbf{A} \circ \mathbf{J}$ ?
- State the Schur product theorem.
- True or false: if  $\mathbf{A} \succ 0$ , then the matrix with entries  $1/a_{ij}$  (assuming no entry is 0) is positive definite. Justify your answer.
- What does Oppenheim's inequality say when  $\mathbf{B} = \mathbf{I}$ ?

*Solution.*

- For  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$  of the **same** shape,  $(\mathbf{A} \circ \mathbf{B})_{ij} = a_{ij}b_{ij}$ , and  $\mathbf{A} \circ \mathbf{B}$  has that same shape ([Definition 12.7.1](#)).
- $\mathbf{A} \circ \mathbf{I} = \text{diag}(a_{11}, \dots, a_{nn})$ , and  $\mathbf{A} \circ \mathbf{J} = \mathbf{A}$ , since  $\mathbf{J}$  has every entry 1.
- If  $\mathbf{A} \succeq 0$  and  $\mathbf{B} \succeq 0$ , then  $\mathbf{A} \circ \mathbf{B} \succeq 0$ ; if both are positive definite, so is  $\mathbf{A} \circ \mathbf{B}$  ([Theorem 12.7.2](#)).
- False. For  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \succ 0$ , the entrywise reciprocal is  $\begin{pmatrix} 1/2 & 1 \\ 1 & 1/2 \end{pmatrix}$ , with determinant

$$-3/4 < 0.$$

- (e) By [Theorem 12.7.3](#) it says  $\det(\mathbf{A} \circ \mathbf{I}) \geq \det \mathbf{A} \cdot 1$ , that is  $a_{11} \cdots a_{nn} \geq \det \mathbf{A}$ : Hadamard's inequality, Theorem 12.6.4 (a).

### 12.7.5.2 B. Practice

#### Exercise B1: Compute and decide.

For each pair below, compute  $\mathbf{A} \circ \mathbf{B}$  and determine whether it is positive definite, positive semidefinite but not definite, or neither. Say in each case whether [Theorem 12.7.2](#) applies.

- (a)  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ .  
 (b)  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ .  
 (c)  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}$ .

*Solution.*

- (a)  $\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ . Both factors are  $\mathbf{1}\mathbf{1}^\top \succeq 0$  (Example (Five Positive Matrices) (b)), so [Theorem 12.7.2](#) (a) applies and predicts  $\succeq 0$ . The product has determinant 0, so it is positive semidefinite but not definite.
- (b)  $\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ , whose determinant is  $1 - 4 = -3 < 0$ : neither. The theorem does not apply, since  $\mathbf{B}$  is not positive semidefinite (its determinant is  $-3$ ). This is what the hypothesis is for.
- (c)  $\mathbf{A} \succ 0$  (minors 2, 3) and  $\mathbf{B} \succ 0$  (minors 3, 2), so [Theorem 12.7.2](#) (b) applies. Indeed  $\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} 6 & -1 \\ -1 & 2 \end{pmatrix}$ , with minors 6 and 11, is positive definite. [Theorem 12.7.3](#) also checks out:  $11 \geq \det \mathbf{A} \cdot b_{11}b_{22} = 3 \cdot 3 = 9$ .

#### Exercise B2: Entrywise squares.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}$ .

- (a) Show that  $\mathbf{A} \succ 0$ .  
 (b) Compute  $\mathbf{A} \circ \mathbf{A}$ , the matrix of squared entries, and verify that it is positive definite.  
 (c) Explain why (b) needed no computation.

*Solution.*

- (a) The leading principal minors are 1,  $2 - 1 = 1$  and, expanding along the first row,  $1(6 - 4) - 1(3 - 2) + 1(2 - 2) = 1$ . All positive, so  $\mathbf{A} \succ 0$  by Theorem 12.1.5 (d).
- (b)  $\mathbf{A} \circ \mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 4 & 4 \\ 1 & 4 & 9 \end{pmatrix}$ , with leading principal minors 1,  $4 - 1 = 3$  and  $1(36 - 16) - 1(9 - 4) + 1(4 - 4) = 20 - 5 = 15$ . All positive, so it is positive definite.
- (c) [Theorem 12.7.2](#) (b) with  $\mathbf{B} = \mathbf{A}$  gives  $\mathbf{A} \circ \mathbf{A} \succ 0$  immediately.

### 12.7.5.3 C. Going deeper



**Exercise C1: Polynomials applied entrywise.**

Let  $\mathbf{A} \in M_n(F)$  with  $\mathbf{A} \succeq 0$ .

- (a) Prove that the entrywise power  $\mathbf{A}^{\circ m}$ , with  $(i, j)$  entry  $a_{ij}^m$ , is positive semidefinite for every integer  $m \geq 1$ .
- (b) Let  $p(x) = c_0 + c_1x + \cdots + c_dx^d$  have **non-negative real** coefficients. Prove that the matrix  $p^\circ(\mathbf{A})$  with  $(i, j)$  entry  $p(a_{ij})$  is positive semidefinite.
- (c) Give an example of  $\mathbf{A} \succeq 0$  and a polynomial  $p$  with a negative coefficient for which  $p^\circ(\mathbf{A})$  is not positive semidefinite.

*Solution.*

- (a) Induct on  $m$ . For  $m = 1$  the claim is the hypothesis. If  $\mathbf{A}^{\circ m} \succeq 0$ , then  $\mathbf{A}^{\circ(m+1)} = \mathbf{A}^{\circ m} \circ \mathbf{A} \succeq 0$  by Theorem 12.7.2 (a).
- (b) By definition of the entries,  $p^\circ(\mathbf{A}) = c_0\mathbf{J} + c_1\mathbf{A} + c_2\mathbf{A}^{\circ 2} + \cdots + c_d\mathbf{A}^{\circ d}$ , since each side has  $(i, j)$  entry  $\sum_k c_k a_{ij}^k$  (with  $a_{ij}^0 = 1$ , which is the  $(i, j)$  entry of  $\mathbf{J}$ ). Now  $\mathbf{J} = \mathbf{1}\mathbf{1}^\top \succeq 0$  by Example (Five Positive Matrices) (b), each  $\mathbf{A}^{\circ k} \succeq 0$  by (a), and each  $c_k \geq 0$ . A non-negative combination of positive semidefinite matrices is positive semidefinite (Exercise (C2: The cone of positive matrices) (a)).
- (c) Take  $\mathbf{A} = \mathbf{J} \in M_2(\mathbb{R})$ , which is positive semidefinite, and  $p(x) = x - 2$ . Then  $p^\circ(\mathbf{A})$  has every entry  $-1$ , that is  $p^\circ(\mathbf{A}) = -\mathbf{J}$ , and  $\langle -\mathbf{J}\mathbf{e}_1, \mathbf{e}_1 \rangle = -1 < 0$ . So the hypothesis on the coefficients cannot be dropped.

**Exercise C2: The product of the determinants.**

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{A} \succeq 0$  and  $\mathbf{B} \succeq 0$ . Prove that

$$\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{A} \cdot \det \mathbf{B}.$$

Is the inequality ever an equality with both matrices positive definite?

*Solution.* By Theorem 12.7.3,  $\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{A} \cdot \prod_i b_{ii}$ . By Theorem 12.6.4 (a) applied to  $\mathbf{B} \succeq 0$ ,  $\prod_i b_{ii} \geq \det \mathbf{B}$ . Both  $\det \mathbf{A} \geq 0$  and  $\prod_i b_{ii} \geq 0$  (the diagonal entries of a positive semidefinite matrix are  $\geq 0$  by Exercise (C1: Zero trace forces zero) (a)), so multiplying the second inequality by  $\det \mathbf{A} \geq 0$  and chaining,

$$\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{A} \prod_i b_{ii} \geq \det \mathbf{A} \det \mathbf{B}.$$

Yes, equality can occur: take  $\mathbf{A} = \mathbf{B} = \mathbf{I}_n$ , so that  $\mathbf{A} \circ \mathbf{B} = \mathbf{I}_n$  and all three quantities are 1. More generally, take  $\mathbf{A} \succ 0$  diagonal and  $\mathbf{B} = \mathbf{I}_n$ .

**Exercise C3: The rank of an entrywise product.**

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{A} \succeq 0$  and  $\mathbf{B} \succeq 0$ , of ranks  $r$  and  $s$ . Prove that

$$\text{rank}(\mathbf{A} \circ \mathbf{B}) \leq rs.$$

*Hint: how many terms in the proof of Theorem 12.7.2 (a) are actually non-zero?*

*Solution.* Write  $\mathbf{A} = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^*$  and  $\mathbf{B} = \sum_j \mu_j \mathbf{w}_j \mathbf{w}_j^*$  as in the proof of Theorem 12.7.2 (a). The number of non-zero eigenvalues of  $\mathbf{A}$  is  $\text{rank } \mathbf{A} = r$ : diagonalizing,  $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^*$  with  $\mathbf{U}$  invertible, so  $\text{rank } \mathbf{A} = \text{rank } \mathbf{D}$  by Theorem 2.5.9, and  $\text{rank } \mathbf{D}$  counts the non-zero  $\lambda_i$ . Likewise exactly  $s$  of the  $\mu_j$  are non-zero. Dropping the vanishing terms,

$$\mathbf{A} \circ \mathbf{B} = \sum_{\lambda_i \neq 0} \sum_{\mu_j \neq 0} \lambda_i \mu_j \mathbf{z}_{ij} \mathbf{z}_{ij}^*, \quad \mathbf{z}_{ij} = \mathbf{u}_i \circ \mathbf{w}_j,$$

a sum of  $rs$  matrices, each of rank at most 1, since every column of  $\mathbf{z}_{ij} \mathbf{z}_{ij}^*$  lies in  $\text{span}(\mathbf{z}_{ij})$ . The rank of a sum of two matrices is at most the sum of their ranks (Exercise (C1: Rank of a sum)); applying that repeatedly to the  $rs$  summands gives  $\text{rank}(\mathbf{A} \circ \mathbf{B}) \leq rs$ .